INCH-POUND

MIL-DTL-24640/22D
22 August 2002
SUPERSEDING
MIL-C-24640/22C
26 June 1995

DETAIL SPECIFICATION SHEET

CABLE, ELECTRIC, 600 VOLTS, LIGHT-WEIGHT, SEVEN-CONDUCTOR, LOW POWER, TYPE 7XW

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and the issue of MIL-DTL-24640 listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation.

Construction (Watertight)

First - Copper conductor, uncoated in accordance with ASTM B 8, 7 strands (watertight). AWG size and dimensions in accordance with Table I for the applicable type.
Second - Mica glass tape insulation 0.0029 ±.0002 inch (50 percent overlap maximum and minimum overlap of 45 percent). Quality conformance testing shall be in accordance with MIL-W-81044, prior to cabling.
Third - Opaque-pigmented polyimide tape .001 inch (50 percent overlap minimum).
Fourth - Opaque-pigmented polyimide tape .001 inch with 0.0001 inch Fluorinated Ethylene Propylene [FEP] (50 percent overlap minimum).
Fifth - Modified aromatic polyimide resin coating 0.0005 inch minimum (.001 inch nominal). Identification codes applied using Method 1 or 3.
Sixth - Seven component wires shall be cabled together. Cabling shall be in sequence starting with number 1 from center outward. Fillers may be used as necessary, to form a round cable.
Seventh - Binder tape applied helically with overlap.
Eighth - Cross-linked polyolefin jacket. (See Table I for thickness).

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

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TABLE I. Construction details.

PIN M24640/20	Type number	AWG size	Conductor diameter (inch) (nom)	Component wire diameter (inch) (min)	(max)	Jacket thickness (inch) (min)	Cable diameter (inch) (min)	(max)	Cable weight (pounds per 1000 feet) (max)	Conductor resistance per 1000 feet (ohms)
-01UN	7XW-3	16	0.058	0.075	0.081	0.038	0.315	0.339	110.0	4.35
-02UN	7XW-4	14	0.073	0.091	0.097	0.041	0.374	0.404	162.0	2.88

REQUIREMENTS:

Qualification required.
Construction details: See Table I.

Component wire: For polyimide tape constructions, the following additional tests shall be applicable in accordance with MIL-W-81381:

Temperature rating: 150°C (302°F) maximum conductor temperature
Voltage rating: 600 volts (rms) at sea level.
Blocking: Oven temperature, 200 ± 2°C (392 ± 3.6°F) no sticking of conductors.
Flammability: 3 seconds (max) after flame.
3.0 inches (max) flame travel.
No flaming of tissue paper.
Humidity resistance: 5 megohms for 1000 feet, minimum insulation resistance after humidity exposure.
Impulse dielectric test: 100 percent test. Impulse voltage at 8.0 kV.
Insulation resistance: 2500 megohms for 1000 feet (min).
Polyimide cure test: No cracking.
Shrinkage: 0.062 inch maximum at 180°C.
Surface resistance: 5 megohms inches (min), initial and final readings.
Thermal shock: Oven temperature, 150± 2°C (302 ± 3.6°F).
Maximum change in measurement: 0.031 inch.
Wet dielectric test: 2500 volts (rms).

EXAMINATION AND TESTS:

Basic electrical: Requirements:
Conductor Resistance - ohms/1000 feet at 25°C, maximum.....See Table I
Voltage withstand - volts, root mean square, minimum
Conductor to conductor 2000
Conductor to shield.....Not applicable
Shield to shield .....Not applicable
Insulation resistance - megohms-1000 feet, minimum.....2500
Conductor and shield continuity.....No failure
Jacket flaws.....Not applicable

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Group A:

- Visual and dimensional ..... No failure
- Capacitance:
  - Mutual capacitance - at 1 MHz, picofarads/foot (pF/ft), maximum ..... Not applicable
  - Capacitance unbalance - percent, maximum ..... Not applicable
- Characteristic impedance - at 1 MHz, ohms ..... Not applicable
- Watertightness (see Table IX of MIL-DTL-24640 for limits of water leakage) ..... No failure
- Weight ..... No failure

Group B:

- Cold bending ..... No failure
- Drip ..... No failure
- Cross link proof test (percent maximum)
  - Jacket 50
- Tear - pounds per inch thickness, minimum 35
  - Physicals (unaged)
    - Jacket
      - Tensile strength - lb/in^2, minimum 1300
      - Elongation - percent, minimum 160
- Attenuation - at 3 MHz, dB/100 feet, maximum ..... Not applicable
- Gas flame (1 hour) ..... No failure

Group C:

- Physicals (aged)
  - Jacket
    - Air oven
      - Tensile strength - percent of unaged, minimum 60
      - Elongation - percent of unaged, minimum 60
- Permanence of printing (conductor) - method 1 only cycles, minimum 25
- Permanence of printing (jacket) - cycles, minimum 125
- Shield - conformance to material, construction and coverage ..... Not applicable
- Heat distortion - percent of unaged, maximum 30
- Cable sealant removability ..... No failure
- Shrinkage ..... No failure
- Mutual inductance - dB/1000 feet, maximum ..... Not applicable
- Surface transfer impedance
  - Milliohms per meter, maximum ..... Not applicable
  - EMP response, dB, minimum ..... Not applicable
- Pulse response time - ms/500 feet
  - Rise time, maximum ..... Not applicable
  - Decay time, maximum ..... Not applicable

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QUALIFICATION INSPECTION:

Qualification inspection shall include basic electrical, all of groups A, B, C and D, plus the following:

Aging and compatibility (cable)	No failure
Acid gas equivalent - percent, maximum	
Jacket	2
Fillers	2
Insulation	18
Halogen content - percent, maximum	
Jacket	0.2
Fillers	0.2
Hot oil immersion	
Tensile strength - percent of unaged, minimum	50
Elongation - percent of unaged, minimum	50
Smoke index, maximum	
Jacket	25
Fillers	45
Insulation	10
Toxicity index, maximum	
Jacket	5
Fillers	5
Insulation	1.5
Durometer (jacket) - (type A) hardness, minimum	80
Weathering (jacket)	No failure
Fungus Resistance	No failure
Abrasion Resistance	No failure
Gas Flame (3 hour)	No failure
Component wire	No failure

Unit ordering length: 500 feet nominal.

Custodians:
Army - MI
Navy - SH

Preparing Activity:
Navy - SH
(Project 6145-2307-022)

Review Activities:
Army - AR, AV, CR, CR4
Navy - CG, EC
DLA - IS

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